

Disagreement-Gated Recertification and Boundary Audits for Anonymous DHT Routers

Keeping certified anonymity–latency menus valid under churn

February 28, 2026 (archive v134)

Abstract

Anonymous DHT defenses are rarely static: operators change retry rules, cover-traffic rates, and thresholded “attack-mode” triggers as churn and adversary pressure shift. Re-certifying every tweak via fresh, fully labeled privacy audits is often prohibitive.

We give a practical recertification protocol that preserves the many-testing-safe guarantees of Certified A while spending audit effort only where the new policy differs from the old. The core device is a *zero-label disagreement gate*: if the new policy’s disagreement rate with a previously certified policy is below the old certificate margin, then the update is immediately safe. If not, we show how to label only the disagreement region and how to concentrate boundary audits where tail/stop-time constraints are tight. A small DHT-shaped toy table (kept in- $\text{\TeX}$) illustrates audit savings.

Scope. We address *recertification under policy drift*: how to keep certified anonymity/latency menus valid when operators tune retries, thresholds, and cover parameters.

Imports. We import certified-menu objects and the baseline many-testing guarantees from Certified A, and routing-signature quantizers from Certified B.[17, 20] We also import stop-time / tail-normal-form background from the published wiki links referenced by Certified A. Exposure-normal-form vocabulary (what counts as the *same* transcript interface) is imported from the trace-interfaces note.[31] Recertification outputs are contract objects (Synthesis 0) and should be anchored by ABOM/OINL manifests when published.[34, 35] Receipt line-item claim fields (including `tw_id` and optional `state_decl_id`) are standardized in Synthesis 12; replay/state drift rules are centralized in Synthesis 18.[32, 33]

Exports. A disagreement-gated recertification protocol: a zero-label gate, disagreement-region auditing rules, and a boundary-audit schedule that preserves Certified A style simultaneous guarantees under drift.

Artifact policy. Source-only; full scripts and raw sizing curves are available on request.[19]

1 Problem: continuous tuning in Anonymous DHTs

Anonymous DHT deployments rarely hold a routing policy fixed. Operators adjust retry caps, cover ratios, and thresholded “attack-mode” branches as churn, load, and adversary pressure shift. Certified A defines *certified menus* (and many-testing-safe certificates) for a baseline policy π_0 ; this

paper addresses the *recertification* problem: given a proposed update π_1 , when can we accept it *without* paying for a full new labeled privacy audit?

Our guiding principle is to spend work only where the update changes behavior. We contribute: (i) a *zero-label disagreement gate* that accepts an update when its disagreement rate with π_0 is below the old certificate margin; (ii) a targeted procedure that labels only the disagreement region when the gate fails; and (iii) *boundary audits* that concentrate effort near tight tail/stop-time constraints. Throughout, we treat “policy” broadly: it may be a concrete router, a choice from a certified menu, or a dial-sheet instantiation of a budgeted knob set. For DHT background, menu objects, and the baseline certification story, see Certified A [17].

2 Setup and notation

Fix an SLA witness W_π encoded as a mean inequality $\mathbb{E}[W_\pi] \leq 0$ as in Certified A. Assume the witness is bounded and, for the update pair (π_0, π_1) , satisfies $|W_{\pi_1} - W_{\pi_0}| \leq 1$ (true for Bernoulli-offset SLAs like privacy, coverage, and tail indicators, and for bounded losses in $[0, 1]$). Let π_0 be a currently deployed policy with a certificate margin:

$$\mathbb{E}[W_{\pi_0}] \leq -\gamma \quad (\gamma > 0). \quad (1)$$

We consider a proposed update π_1 and want to decide whether (and how) to recertify it.

3 First question: is this the same claim?

Before invoking any disagreement gate, a deployment must decide whether the update preserves the *certified interface*. Certified A certificates and the gate below are statements about a declared witness surface, repetition unit, and public conditioning. If those move, the update is not merely “a new policy on the same claim”; it may be a *different claim object*. [31, 34]

Receipt-level identity check. In this archive, “same claim” is intended to be mechanical: the public receipt tuple includes `exposure_nf_id` (the witness interface), `tw_id` (the repetition window/threat declaration), and optionally `state_decl_id` (the state surface assumed). If any of these ids changes, the safe default is to mint a new claim object (and then decide whether it is a conservative coarsening of a prior interface). [32, 33]

Lemma 1 (Interface-preserving drift is a prerequisite for the gate). *Consider a baseline certificate for policy π_0 at exposure normal form $(\mathcal{Y}, g, \text{unit}, \text{cond})$. A recertification shortcut based on disagreement between g_{π_0} and g_{π_1} is sound only if one of the following holds:*

1. *the update keeps the same exposure normal form up to metadata/version labels; or*
2. *the updated certificate explicitly coarsens the old interface, i.e. the new observation is $g' = h \circ g$ under the same repetition unit and public conditioning.*

If the update changes the observation map in any other way, or changes the repetition unit or public conditioning, then the correct action is to mint a new claim object and certify that new interface directly.

Proof sketch. The disagreement argument compares two policies on the *same experiment*: the same witness extractor, composed over the same repetition unit and under the same public conditioning. If the interface changes only by an explicit coarsening, the certified endpoint can only improve by data processing, so reuse is conservative. Otherwise the witness channel itself changes, and the original accountant no longer certifies the updated claim. $\square$

Operational rule. Run a “same exposure normal form?” check *before* the zero-label gate. Only after the interface is fixed should a deployment spend unlabeled evidence on disagreement estimation.

Aside: replay-plan drift. If the privacy claim is unchanged but the replay workflow changes (`plan_spec_id` drift), the publication must still republish a new receipt/ABOM pinning the updated replay promise. Whether that drift triggers new privacy evidence is classified by the P0/P1/P2 taxonomy in Synthesis 18.[33]

4 Zero-label disagreement gate

Let $g_\pi(X)$ denote the *routing decision* induced by policy π in context X (which template is used; which threshold branch; ...). For randomized policies, treat the per-episode PRG seed U as part of the context; see Certified B [20]. In implementation, it is often simplest to set $g_\pi(X) := Q(\pi(X))$ for the same quantizer Q used in routing-signature compression (Certified B), so that disagreement is measured on *realized* decisions. In particular, g_π should include any action components that can affect an SLA witness (padding schedule, retry count, dummy rate), not just a coarse “mode bit.” Define the *disagreement rate*

$$q := \mathbb{P}(g_{\pi_0}(X) \neq g_{\pi_1}(X)),$$

estimable from unlabeled logs (contexts only). With a finite unlabeled log of size n_u , we estimate q by $\hat{q}$ (the empirical disagreement rate) and form a one-sided upper confidence bound $\bar{q}$. As in Certified A, compute a one-sided upper confidence bound $\bar{q}$ on q from unlabeled contexts (e.g., the exact Clopper–Pearson upper bound) [4]. This is the same structural object that governs disagreement-based active learning (the “disagreement region”), though our goal is certification rather than learning [16].

Closed form for OR-threshold routers (uniform context model). If $X = (a, c)$ is uniform on $[0, 1]^2$ and $g_\pi(a, c) = \mathbf{1}\{a > \theta_a \text{ or } c > \theta_c\}$, then $g_\pi = 0$ only on the rectangle $[0, \theta_a] \times [0, \theta_c]$. The disagreement rate between $\pi_0 = (\theta_a, \theta_c)$ and $\pi_1 = (\theta'_a, \theta'_c)$ is the symmetric-difference area of these rectangles:

$$q = \theta_a \theta_c + \theta'_a \theta'_c - 2 \min(\theta_a, \theta'_a) \min(\theta_c, \theta'_c). \quad (2)$$

In practice q is estimated from unlabeled logs under the deployment context distribution (not necessarily uniform), but the rectangle formula is useful for sanity checks and for designing threshold step sizes that stay within a desired disagreement budget.

We impose a mild condition that holds for most thresholded anonymous routing designs:

Whenever two policies take the same routing decision on a context, their SLA witness is identical on that episode. Formally, $W_{\pi_0}(X, Y) = W_{\pi_1}(X, Y)$ whenever $g_{\pi_0}(X) = g_{\pi_1}(X)$.

Lemma 2 (Zero-label safety check from disagreement). *Assume $W_{\pi_0}(X, Y) = W_{\pi_1}(X, Y)$ whenever $g_{\pi_0}(X) = g_{\pi_1}(X)$ and $|W_{\pi_1} - W_{\pi_0}| \leq 1$. If π_0 satisfies (1), then*

$$\mathbb{E}[W_{\pi_1}] \leq -\gamma + q.$$

In particular, if $q \leq \gamma$, then $\mathbb{E}[W_{\pi_1}] \leq 0$ and the update is safe without any new labels.

Remark 1 (Multiple SLAs: gate by the tightest margin). *If Certified A certified K SLA witnesses with margins $\gamma_1, \dots, \gamma_K$ for π_0 , then the same disagreement rate q upper-bounds the change for each bounded witness. Thus a sufficient zero-label condition to preserve all SLAs is $q \leq \min_k \gamma_k$ (apply Theorem 1 with a union bound over the K gates). If only a subset of SLAs is near-tight, it is often effective to gate on those and immediately trigger targeted audits for the rest.*

Interpretation. If the new policy disagrees with the old only rarely (under the context distribution), then the old certificate “pays” for the change.

Relation to safe policy improvement. The gate is a DHT-native analogue of *safe policy improvement*: we only accept an update when we can certify it does not violate a previously certified baseline constraint. In reinforcement-learning and bandit deployments, closely related ideas accept policy updates based on high-confidence off-policy evaluation and baseline guarantees [10, 11]. Our setting differs in that the witness is a per-episode, bounded SLA functional (privacy/coverage/tail/cost) and we exploit the discrete routing structure to localize audits to a disagreement region.

Theorem 1 (Gate with estimated disagreement (finite-sample)). *Assume π_0 has been certified so that (1) holds with probability at least $1 - \delta_0$ under the deployment distribution. Let $\bar{q}$ be any upper confidence bound on the disagreement rate q computed from an independent unlabeled context log, such that $\mathbb{P}(q \leq \bar{q}) \geq 1 - \delta_1$ (e.g., Clopper–Pearson inversion). If the update is accepted only when $\bar{q} \leq \gamma$, then with probability at least $1 - (\delta_0 + \delta_1)$, the accepted update satisfies $\mathbb{E}[W_{\pi_1}] \leq 0$.*

5 When the gate fails: label only the disagreement region

If $q > \gamma/2$, we should not blindly deploy, but we still do not need to label everything. Let $\mathcal{R} = \{X : g_{\pi_0}(X) \neq g_{\pi_1}(X)\}$ be the disagreement region. Only episodes with $X \in \mathcal{R}$ can change the SLA witness.

Writing $\Delta W := W_{\pi_1} - W_{\pi_0}$, note that $\Delta W = 0$ off $\mathcal{R}$, so

$$\mathbb{E}[W_{\pi_1}] = \mathbb{E}[W_{\pi_0}] + q \mathbb{E}[\Delta W \mid X \in \mathcal{R}].$$

Thus, when the gate fails ($q > \gamma$), it suffices to label only $X \in \mathcal{R}$ and obtain a one-sided upper bound on $\mathbb{E}[\Delta W \mid X \in \mathcal{R}]$, yielding a tight incremental certificate.

Theorem 2 (Incremental certificate from disagreement-band labels). *Assume π_0 satisfies (1) with probability at least $1 - \delta_0$. Let $\bar{q}$ be an upper confidence bound on q from an independent unlabeled log such that $\mathbb{P}(q \leq \bar{q}) \geq 1 - \delta_1$. Let $\bar{\Delta}$ be an upper confidence bound on the conditional mean $\mathbb{E}[\Delta W \mid X \in \mathcal{R}]$ computed from (possibly selectively sampled) labeled data on $\mathcal{R}$ with known propensities (e.g., via Horvitz–Thompson / IPS weighting), such that $\mathbb{P}(\mathbb{E}[\Delta W \mid X \in \mathcal{R}] \leq \bar{\Delta}) \geq 1 - \delta_2$. Then with probability at least $1 - (\delta_0 + \delta_1 + \delta_2)$,*

$$\mathbb{E}[W_{\pi_1}] \leq -\gamma + \bar{q} \bar{\Delta}.$$

In particular, if $\bar{q} \bar{\Delta} \leq \gamma$ then the update is certified-safe.

Importance correction for selective labeling. If we only label a subset of episodes (e.g., only those with $X \in \mathcal{R}$), the labeled sample is biased. A clean deployment pattern is to assign a known labeling propensity $\rho(X) \in (0, 1]$ concentrated on $\mathcal{R}$ (e.g., $\rho(X) = \rho_0$ on $\mathcal{R}$ and a small floor $\rho_{\min} > 0$ elsewhere if unconditional estimates are needed), and apply inverse-probability weighting. Let $L_i \sim \text{Bernoulli}(\rho(X_i))$ be the label-indicator and let $\widetilde{W}_i$ be the witness computed when the label is obtained. Then the Horvitz–Thompson / IPS estimator

$$\widehat{\mathbb{E}}[W_{\pi_1}] := \frac{1}{n} \sum_{i=1}^n \frac{L_i \widetilde{W}_i}{\rho(X_i)}$$

Algorithm 1 Disagreement-gated recertification (single SLA)

Require: Certified policy π_0 with margin γ ; proposed update π_1 ; unlabeled contexts; overall failure δ .

- 1: Estimate $\hat{q}$ from contexts and form an upper confidence bound $\bar{q} = \text{CP_UCB}(\hat{q}; n_u, \delta)$ using the one-sided Clopper–Pearson (exact binomial) bound, where n_u is the unlabeled context sample size.
 - 2: **if** $\bar{q} \leq \gamma$ **then**
 - 3: **return** **Accept** update (zero labels).
 - 4: **else**
 - 5: Collect/label episodes with a known propensity $\rho(X)$ concentrated on $\hat{\mathcal{R}}$ (disagreement band).
 - 6: Certify $\mathbb{E}[W_{\pi_1}] \leq 0$ using IPS/HT weighting on labeled data, with Bonferroni/Holm correction as needed.
 - 7: **return** **Accept** only if certified.
 - 8: **end if**
-

is unbiased under mild conditions [9]. If the deployment monitors or re-audits continuously, the statistical meaning should survive optional stopping and repeated publication. For a receipt-friendly (union-bound) way to publish a single monitoring confidence budget and “spend” it over repeated audit events, see Synthesis 26.[18]

A minimal recertification protocol:

1. estimate q from unlabeled contexts;
2. if $q \leq \gamma$, accept (Lemma 2);
3. else, selectively label episodes in $\mathcal{R}$ and certify $\mathbb{E}[W_{\pi_1} \mid X \in \mathcal{R}]$ with an importance correction.

Implementation note. $\text{CP_UCB}(\hat{q}; n_u, \delta)$ is the smallest u such that $\Pr(\text{Bin}(n_u, u) \leq n_u \hat{q}) \geq 1 - \delta$.

Remark 2 (Multiple SLAs and many-testing). *To maintain Certified A’s guarantees under updates, apply Bonferroni/Holm across all constraints and all candidate updates tested. Disagreement gating reduces the data needed per update, not the need for correct multiple-testing control.*

6 Boundary audits for tail and stop-time SLAs

Stop-time hiding and tail-latency guarantees are typically certified via rare-event indicators [22, 21]. Naively estimating rare-event probabilities can require enormous samples. However, anonymous DHT routers often produce a *score* $s(X)$ (congestion/attack score) and implement policies as thresholds in s . This induces a *boundary* where violations concentrate.

A practical scheme:

- bin contexts by score (or by proximity to policy thresholds);
- oversample and label more heavily near the boundary where tail events occur;
- combine bin-wise estimates into an overall tail certificate.

Remark 3 (How to allocate boundary-audit labels across slices). *If contexts are stratified into slices h with mass p_h (from unlabeled telemetry) and within-slice standard deviation s_h for the witness*

threshold update (θ_a, θ_c)	disagreement q	gate (margin $\gamma = 0.05$)	labels if audit needed
$(0.60, 0.60) \rightarrow (0.61, 0.59)$	0.0175	pass	0
$(0.60, 0.60) \rightarrow (0.65, 0.65)$	0.0817	fail	981
$(0.60, 0.60) \rightarrow (0.40, 0.40)$	0.3618	fail	4342

Table 1: Disagreement gating for OR-threshold routers using an unlabeled context log (here drawn from Certified A’s simulator). For the decision rule “high iff $a > \theta_a$ or $c > \theta_c$,” disagreement is the symmetric-difference area of two rectangles and can be computed exactly (Equation (2)). If an update fails the gate, auditing only the disagreement region reduces labels from a full-sample audit to about q times that amount.

being certified, then under a fixed total label budget n the variance of the stratified mean estimator is minimized by allocating labels as $n_h \propto p_h s_h$ (Neyman/optimum allocation) [1, 2]. In practice, use a pilot batch to estimate s_h and round n_h to integers.

Off-policy and shadow-routing for audits. When the disagreement region is small, it can be efficient to log a small randomized exploration floor (shadow-routing) so that disagreement-band sampling is not biased. The resulting audit is an off-policy evaluation problem; see Certified A for background, references, and high-confidence off-policy certification pointers.

7 Simulator evidence (kept in- $\text{\LaTeX}$)

We include two small DHT-shaped tables to validate the workflow end-to-end. As in Certified A, these are not claimed as a full Octopus/Salsa/ShadowWalker simulator; they exist to make the certification and recertification protocol concrete.

7.1 Disagreement gate: updates accepted with zero new labels

We consider an OR-threshold anonymous router that chooses between two routing templates based on thresholds (θ_a, θ_c) on an “attack pressure” score a and “congestion” score c . We estimate the disagreement rate q from unlabeled contexts (here from Certified A’s simulator context distribution); Equation (2) gives a closed form under a uniform model as a sanity check.

Table 1 shows three representative updates for a policy with privacy margin $\gamma = 0.05$. A small tweak passes the gate (no new labels); larger threshold changes fail the gate and trigger disagreement-band auditing. The last column illustrates a simple label-budget heuristic: if a full-sample audit uses $n = 12,000$ episodes, auditing only the disagreement region costs about qn labels.

7.2 Boundary audits for rare tail constraints

We illustrate boundary auditing for a rarer stop-time/latency constraint by slicing contexts by congestion quantiles. Using the minimal simulator from Certified A and a rarer tail threshold $t = 1100\text{ms}$, tail violations concentrate in the high-congestion region.

8 Monitoring drift without breaking guarantees

Between discrete recertifications, operators often want an *anytime* alarm if SLAs begin to fail under drift. Time-uniform confidence sequences for means provide such guarantees under minimal

congestion slice	slice mass	tail rate in slice	contribution to total tail
bottom 90%	0.90	0.035778	0.032200
next 7%	0.07	0.209714	0.014680
next 2%	0.02	0.498000	0.009960
top 1%	0.01	0.568000	0.005680

Table 2: Boundary audit illustration from the minimal simulator (Certified A), evaluated on one certified menu point, using a rarer tail threshold $t = 1100\text{ms}$. The unconditional tail rate is 0.0625, while the conditional tail rate in the top 3% congestion mass is 0.521 (about $8.3\times$ larger), so oversampling near the boundary yields many more “tail hits” per label.

assumptions [3] and fit naturally into safe anytime-valid inference workflows [5]. A simple deployment pattern:

- maintain an always-valid upper confidence bound on $\mathbb{E}[W_{\pi_t}]$ from streaming telemetry;
- if the bound crosses 0, trigger disagreement-gated recertification for a new policy.

This prevents optional stopping from invalidating risk monitoring.

Many updates over time. If operators test a stream of policy updates or monitor multiple menu points simultaneously, the relevant multiplicity grows over time. Online family-wise error control (alpha-spending and its refinements) provides a principled way to maintain a global error budget across an unbounded sequence of tests [6]. Equivalently, one can use *e-values* / test supermartingales to obtain optional-stopping-safe alarms and combine evidence across time [8, 15, 7].

Artifacts

Full scripts and raw sizing curves are in the research archive (available on request). This note keeps only the theory and minimal evidence.

9 Takeaways

- **Recertification is an adversarial interface:** changing knobs on demand can amplify many-testing effects unless changes are gated by explicit triggers and accounted for.
- **Disagreement-gated updates:** only permit recertification when monitors/auditors observe statistically significant disagreement with the published budget, and bind triggers to public randomness to prevent cherry-picking.
- **Pair with signature compression:** when large knob grids are redundant, compress to M_{eff} before applying multiplicity corrections; otherwise re-certification becomes overly conservative.
- **Operational moral:** publish what you account (witness projections and contexts) so “budget drift” is detectable without full trace disclosure.

References

- [1] J. Neyman. On the Two Different Aspects of the Representative Method: The Method of Stratified Sampling and the Method of Purposive Selection. *Journal of the Royal Statistical Society*, 97(4):558–606, 1934. DOI: 10.1111/j.2397-2335.1934.tb04184.x.
- [2] W. G. Cochran. *Sampling Techniques* (3rd ed.). John Wiley & Sons, 1977.
- [3] S. R. Howard, A. Ramdas, J. McAuliffe, and J. Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics*, 49(2):1051–1080, 2021.
- [4] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. DOI: 10.1093/biomet/26.4.404.
- [5] A. Ramdas, P. Grünwald, V. Vovk, and G. Shafer. Game-Theoretic Statistics and Safe Anytime-Valid Inference. *Statistical Science*, 38(4):576–597, 2023. DOI: 10.1214/23-STS894. (Also arXiv:2210.01948.)
- [6] J. Tian and A. Ramdas. Online control of the familywise error rate. *Statistical Methods in Medical Research*, 30(4):976–996, 2021. (Also arXiv:1910.04900.)
- [7] W. Hartog and L. Lei. Family-wise Error Rate Control with E-values. *arXiv:2501.09015*, 2025. DOI: 10.48550/arXiv.2501.09015.
- [8] A. Ramdas and R. Wang. Hypothesis testing with e-values. *Foundations and Trends in Statistics* (monograph); arXiv:2410.23614v6, 2025. DOI: 10.48550/arXiv.2410.23614.
- [9] D. G. Horvitz and D. J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952. DOI: 10.1080/01621459.1952.10483446.
- [10] P. S. Thomas, G. Theodoropoulos, and M. Ghavamzadeh. High-Confidence Off-Policy Evaluation. *AAAI Conference on Artificial Intelligence*, 2015, pp. 3000–3006. <https://people.cs.umass.edu/~pthomas/papers/Thomas2015.pdf>.
- [11] R. Laroche, P. Trichelair, and R. Tachet des Combes. Safe Policy Improvement with Baseline Bootstrapping. *ICML (PMLR 97)*, 2019. <https://proceedings.mlr.press/v97/laroche19a/laroche19a.pdf>.
- [12] S. Sridhar, O. Ascigil, N. V. Keizer, F. Genon, S. Pierre, Y. Psaras, E. Rivière, and M. Król. Content Censorship in the InterPlanetary File System. *Network and Distributed System Security Symposium (NDSS)*, 2024. DOI: 10.14722/ndss.2024.23153. (Also arXiv:2307.12212.) <https://www.ndss-symposium.org/ndss-paper/content-censorship-in-the-interplanetary-file-system/>.
- [13] T. Cholez and C.-L. Ignat. Sybil Attack Strikes Again: Denying Content Access in IPFS with a Single Computer. *Proceedings of the 19th International Conference on Availability, Reliability and Security (ARES)*, 2024. DOI: 10.1145/3664476.3664482. <https://dl.acm.org/doi/10.1145/3664476.3664482>.
- [14] V. H. M. Netto, T. Cholez, and C.-L. Ignat. Active Sybil Attack and Efficient Defense Strategy in IPFS DHT. *arXiv:2505.01139*, 2025. DOI: 10.48550/arXiv.2505.01139.
- [15] V. Vovk and R. Wang. E-values: Calibration, combination, and applications. *Annals of Statistics*, 49(3):1736–1754, 2021. DOI: 10.1214/20-AOS2020. (arXiv:1912.06116).
- [16] S. Hanneke. Theory of Disagreement-Based Active Learning. *Foundations and Trends in Machine Learning*, 7(2–3):131–309, 2014. DOI: 10.1561/22000000037.

- [17] Anonymous DHT Series. *Certified Menus for Anonymous DHT Lookups*. (Certified A in this series, draft.)
- [18] Anonymous. Anytime Statistical Spending for Menu Audits: Change-Control-Safe Confidence Budgets for Expanding Log Certificates. Synthesis 26, The-One-True-Archive (2026).
- [19] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [20] Certified Series. *Routing-Signature Compression*. (Series paper in this archive: `certified_series/paperB.routing-signature-compression/paper.tex`.)
- [21] Anonymous DHT Series. *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups*. Wiki: `[[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]`. (Series paper, already published.)
- [22] Anonymous DHT Series. *Stop-Time Padding Addendum (unique-only)*. (Series addendum, already published.)
- [23] J. Matter and M. Tran. Network-level Censorship Attacks in the InterPlanetary File System. *arXiv:2509.06626*, 2025. DOI: 10.48550/arXiv.2509.06626. <https://arxiv.org/abs/2509.06626>.
- [24] B. Prünster, A. Marsalek, and T. Zefferer. Total Eclipse of the Heart – Disrupting the InterPlanetary File System. *31st USENIX Security Symposium (USENIX Security '22)*, 2022. <https://www.usenix.org/system/files/sec22-prunster.pdf>.
- [25] I2P Project. The Network Database (netDb) documentation (floodfill; XOR-metric distance; daily routing-key “keyspace rotation”). <https://beta.i2p.net/en/docs/overview/network-database/>. (Updated Mar. 2025; accurate for 0.9.65; accessed Feb. 2026.)
- [26] I2P Project. Network Database discussion (history; legacy Kademlia fallback disabled). <https://geti2p.net/en/docs/discussions/netdb>. (Accessed Feb. 2026.)
- [27] C. Egger, J. Schlumberger, C. Kruegel, and G. Vigna. Practical Attacks against the I2P Network. *Research in Attacks, Intrusions, and Defenses (RAID)* (LNCS 8145), 2013, pp. 432–451. DOI: 10.1007/978-3-642-41284-4_22. https://link.springer.com/chapter/10.1007/978-3-642-41284-4_22.
- [28] H. Wang, Z. Ling, X. Xu, Y. Pan, G. Liu, J. Luo, and X. Fu. Time will Tell: Large-scale De-anonymization of Hidden I2P Services via Live Behavior Alignment (Extended Version). *arXiv:2512.15510*, Dec. 2025. <https://arxiv.org/abs/2512.15510>.
- [29] H. Wang, Z. Ling, X. Xu, Y. Pan, G. Liu, J. Luo, and X. Fu. Time will Tell: Large-scale De-anonymization of Hidden I2P Services via Live Behavior Alignment. In *Network and Distributed System Security Symposium (NDSS)*, 2026. <https://www.ndss-symposium.org/ndss-paper/time-will-tell-large-scale-de-anonymization-of-hidden-i2p-services-via-live-behavior-align>
- [30] P. Wang, J. Tyra, E. Chan-Tin, T. Malchow, D. Foo Kune, N. Hopper, and Y. Kim. Attacking the Kad network—real world evaluation and high fidelity simulation using DVN. *Security and Communication Networks*, 6(12):1556–1575, 2013. DOI: 10.1002/sec.172.
- [31] Anonymous. Trace Interfaces for Anonymous DHTs: Observation Tiers, Committee-Contact Traces, and Exposure Normal Forms. Synthesis 3, The-One-True-Archive (2026).
- [32] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).

- [33] Anonymous. Digest-Bound Replay Plans and State Declarations: Equivalence, Minimality, and Change Control. Synthesis 18, The-One-True-Archive (2026).
- [34] Anonymous. Contract Objects for Anonymous-DHT Privacy Budgets and Claims. Series synthesis note (Synthesis 0), 2026. (This archive.)
- [35] Anonymous. ABOM/OINL: Audit BOM and Outcome Identification Noise Label for Anonymous DHT Deployments. Anonymity series Paper C, 2026. (This archive.)